

Interaction-Aware Highlight Selection and Caption Generation in Gaming Content

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Abstract

With the rapid growth of short-form video content, the importance of automatically selecting semantically meaningful segments from long videos and generating natural language descriptions has increased. While narratives in game videos are often formed by interactions among temporally clustered events rather than isolated actions, existing video captioning models struggle to capture such higher-level context. We reformulate short-form game video generation as an interaction-aware scene understanding problem and propose a method that models adjacent events as a unified interaction unit. Our approach combines event classification with Kernel Density Estimation (KDE) to automatically identify salient interaction segments and generates contextually coherent captions conditioned on these segments. Experimental results show that the proposed model consistently outperforms the existing video captioning baseline. Moreover, it demonstrates strong applicability to real-world user-recorded gameplay videos without requiring expert commentary or manual editing.

1. Introduction

As the demand for short-form video content continues to grow, the importance of automatically selecting semantically important segments from long videos and describing them with natural language captions has increased (May & Siddoo, 2024). Short-form videos, typically ranging from 15 seconds to one minute, require the ability to compactly convey key events and their progression within a limited time budget. Accordingly, vision-language models have evolved beyond global video summarization toward recognizing and describing events occurring within videos (Li et al., 2018;

Table 1. Task scope of existing video captioning methods. Interaction-aware refers to capturing scene-level semantics that emerge from multiple temporally clustered events.

Method	Domain	Caption Generation	Event Localization	Event Classification	Interaction Awareness
Vid2Seq (Yang et al., 2023)	Open	✓	✓		
CM ² (Kim et al., 2024)	Open	✓	✓		
EgoYC2 (Ohkawa et al., 2025)	Ego	✓	✓		
FSN (Yu et al., 2018)	Sports	✓	✓		
MatchVision (Rao et al., 2025)	Sports	✓		✓	
LoL-V2T (Tanaka & Simo-Serra, 2021)	Game	✓			
Ours	Game	✓	✓	✓	✓

Islam et al., 2024; Zhou et al., 2024; Kim et al., 2024; Barua et al., 2025).

Accordingly, prior work has focused on combining caption generation with event localization (Yang et al., 2023; Kim et al., 2024; Ohkawa et al., 2025; Yu et al., 2018) or event classification (Rao et al., 2025), as summarized in Table 1, to more precisely capture and describe video content. While these approaches enable event-level understanding with temporal context, they are typically designed to generate descriptions at the level of individual and atomic actions. As a result, they struggle to capture higher-level contextual meaning that emerges when multiple temporally adjacent events form a single coherent scene. This limitation is particularly pronounced in the game domain, where narratives emerge from real-time interactions among multiple agents (Tanaka & Simo-Serra, 2021). In practice, short-form game videos are often composed not of isolated events, but of interaction-centric scenes formed by a sequence of events occurring within a short time span. In such cases, the temporal structure, co-occurrence, and collective dynamics of events play a central role in determining the semantic meaning of the scene. Therefore, effective short-form game video generation requires moving beyond event-level recognition toward modeling interactions formed by temporally clustered events as a unified semantic unit.

To address this challenge, we propose *Interaction-Aware Highlight Selection and Caption Generation*, a model for short-form content generation from game videos. Our approach follows a two-stage pipeline consisting of a Salient Interaction Module and a Caption Generation Module (see Figure 1). In the first stage, the Salient Interaction Module detects and classifies semantically important in-game

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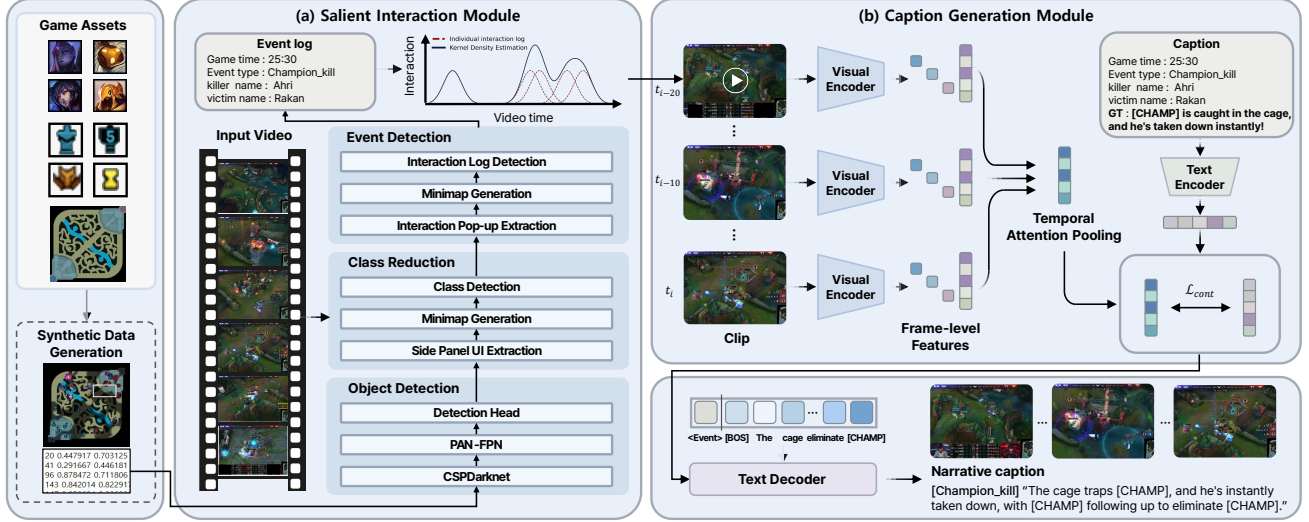


Figure 1. Overview of the proposed Interaction-Aware Highlight Selection and Caption Generation model

events. It then identifies interaction segments based on the temporal density of event occurrences. To account for the non-uniform and clustered temporal distribution of interactions, we employ Kernel Density Estimation (KDE) (Parzen, 1962a; Davis et al., 2011b) to automatically select salient interaction segments. In the second stage, the Caption Generation Module trains a vision-language model conditioned on events within the selected interaction segments and generates coherent narrative captions that describe scenes at the interaction level. Experimental results demonstrate that the proposed model consistently outperforms the existing video captioning baseline. Moreover, our approach can generate short-form game content from user-recorded gameplay videos without requiring professional commentary or manual editing, indicating strong applicability in real-world scenarios.

Our main contributions are summarized as follows:

- (i) We reformulate short-form game video generation as an interaction-aware scene understanding problem, in which temporally clustered events are modeled as a unified interaction unit rather than as isolated events.
- (ii) We propose an interaction segmentation method that combines event classification with KDE to automatically identify semantically salient interaction segments.
- (iii) Through extensive experiments, we demonstrate that the proposed model outperforms the existing video captioning baseline and generalizes effectively to real-world gameplay videos.

2. Related works

Dense Video Captioning. Dense video captioning (DVC) aims to temporally localize salient events in a video and generate a natural language description for each detected

event. Early approaches predominantly adopted a two-stage pipeline, where candidate temporal segments are first generated and then described individually. Krishna et al. (Krishna et al., 2017) proposed a two-stage model that decouples event proposal from caption generation, employing Long Short-Term Memory (LSTM) based models to generate captions for each localized event. Subsequent works extended this line of research by improving temporal modeling and reducing the reliance on heuristic proposal generation (Deng et al., 2021; Zhou et al., 2018b; Zhu et al., 2022; Li et al., 2018; Yang et al., 2023). PDVC (Wang et al., 2021) reformulated DVC as a set prediction problem, employing transformer-based encoders and decoders to jointly localize events and generate captions, thereby eliminating the need for an explicit proposal stage. Similarly, Vid2Seq (Yang et al., 2023) adopted a unified sequence-to-sequence formulation, leveraging large-scale pretraining to jointly model event localization and caption generation in an end-to-end manner. More recently, Large Language Model (LLM)-based approaches have been explored to further improve DVC by leveraging pretrained large language models to enhance linguistic expressiveness in generated descriptions (Wu et al., 2024; Li et al., 2024). Most existing approaches are evaluated on general-purpose benchmarks, including ActivityNet Captions (Krishna et al., 2017), YouCook2 (Zhou et al., 2018a), MSR-VTT (Xu et al., 2016), and Ego4D (Grauman et al., 2022), and primarily focus on improving caption quality for general videos.

Video Highlight Detection. Video highlight detection and event localization focused on identifying important moments from long videos (Xiong et al., 2019). These tasks have been studied using a variety of approaches, including fully supervised, weakly supervised, and unsupervised methods (Badamdorj et al., 2022; Islam et al., 2025; Cai et al.,

Table 2. Comparison of existing video-language datasets. Our dataset significantly surpasses existing video game datasets in both the number of videos and total duration. *Narration* indicates whether sentences are derived from spoken narration within the video.

Dataset	Domain	Videos			Sentences			Narration
		Num. Videos	Total Duration	Duration / Video	Num. Sentences	Total Tokens	Words / Sentence	
MSR-VTT (Xu et al., 2016)	Open	7.2k	40h	0.33min	200k	1.86M	9.3	
ActivityNet-Caption (Krishna et al., 2017)	Open	20k	849h	2.55min	100k	1.35M	13.48	
YouCook2 (Zhou et al., 2018a)	Cooking	2k	176h	5.28min	15.4k	0.14M	8.8	
TACOS (Das et al., 2013)	Cooking	127	15.9h	7.51min	18.2k	0.2M	10.5	
TACOS-ML (Rohrbach et al., 2014)	Cooking	185	27.1h	8.79min	52.5k	—	—	
Ego4D (Grauman et al., 2022)	Ego	9.6k	3670h	22.94min	3.85M	—	—	
DiDeMo (Anne Hendricks et al., 2017)	Open	10k	88.7h	0.53min	40.5k	0.3M	8.0	
MAD (Soldan et al., 2022)	Movie	650	1207.3h	111.44min	384.6k	4.9M	12.7	✓
SoccerReplay-1988 (Rao et al., 2025)	Sports	1.99k	3323h	100.3min	150k	—	—	✓
Getting Over It (Li et al., 2019)	Video Game	8	1.8h	13.50min	2.27k	—	—	✓
LoL-V2T (Tanaka & Simo-Serra, 2021)	Video Game	157	76h	29.05min	63k	0.97M	15.4	✓
Game-MUG (Zhang et al., 2024)	Video Game	216	—	—	70.7k	—	—	✓
Ours	Video Game	1.17k	696.27h	35.58min	31.16k	0.68M	21.8	✓

2018; Chu et al., 2015; Wang et al., 2020; Rochan et al., 2020). Most existing studies were conducted on general-purpose datasets, such as cooking videos, sports videos, and benchmarks like TVSum (Lei et al., 2021; Song et al., 2015; Yao et al., 2016). With the growth of the gaming industry, highlight detection has also been studied in the context of game streaming and esports broadcasts (Chu & Chou, 2015a; Ringer & Nicolaou, 2018; Chu & Chou, 2017). (Chu & Chou, 2015b) adapted highlight detection by jointly analyzing in-game event messages, visual scene changes, and viewer responses such as chat activity to detect highlights in League of Legends (LoL) broadcast videos. (Ringer et al., 2022) proposed an approach that assigned importance scores to fixed-length temporal segments based on multimodal features, including visual, audio, and textual cues, to identify highlights within a video. Despite these efforts, highlight detection in gaming videos remained challenging. Gaming videos exhibited frequent and rapid visual changes even outside of highlight moments, which limited the effectiveness of visual-saliency-based methods. Moreover, when multiple events occurred in close temporal proximity, on-screen textual cues in game broadcasts were displayed sequentially, causing the text to appear later than the actual event times and making accurate temporal highlight localization difficult.

Game Video Commentary Generation. Game videos exhibit visual structures and information delivery mechanisms that differ fundamentally from those of general videos such as open-world footage or sports broadcasts. Game state information, including health indicators and minimap-based positional cues, is explicitly visualized through the heads-up display (HUD) in real time. In addition, event logs, such as player eliminations or neutral object secures, are rendered on screen and serve as a primary source of semantic information for game video understanding (Wang & Yoshinaga, 2024b;a; Zhang et al., 2024). In contrast, general videos lack

explicit state information such as an actor’s health, location, or goal completion status, forcing interpretation to rely heavily on subjective inference. Game videos instead represent multi-agent systems governed by well-defined rules, where identical state information is consistently available to all viewers (Zhang et al., 2022). Consequently, game video commentary tends to focus more on strategic context and player intent than on purely perceptual changes.

Leveraging these properties, (Wang & Yoshinaga, 2024b) formulate commentary generation as a data-to-text task by mapping structured sequences of game events to natural language commentary. Similarly, (Zhang et al., 2024) study multimodal game situation understanding by combining game event logs, caster commentary, and audience chat signals. Despite their effectiveness, these approaches share a common limitation. They rely on structured game data provided by the game system, rather than recognizing events directly from the visual content of the game screen. (Tanaka & Simo-Serra, 2021) segment full match videos into clips using automatic scene transition detection and ResNet-50-based filtering (Xie et al., 2017), followed by clip-level caption generation. However, such approaches often fail to capture the temporal continuity of events and interactions, resulting in captions that lack coherent contextual flow across clips. As a result, existing approaches fail to adequately capture semantic structures formed by interaction-level compositions of multiple events, which collectively constitute highlight-worthy moments in game videos. Therefore, our work directly detects on-screen game event logs and identifies interaction segments based on the temporal density of event occurrences. By organizing these segments into interaction-level semantic units, we aim to generate captions that more faithfully reflect the contextual dynamics of gameplay.

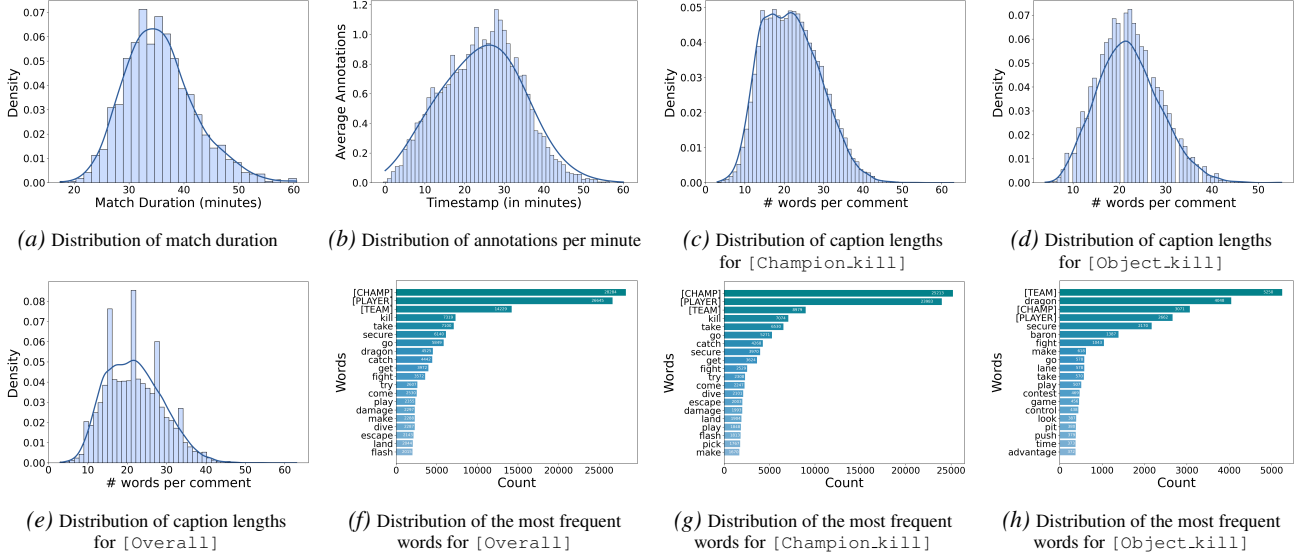


Figure 2. Statistics of the commentary dataset

3. Dataset

Existing game video datasets are relatively limited in scale compared to video datasets from other domains, as summarized in Table 2. This limitation becomes more pronounced in the context of game broadcasts, where utterances tend to be longer and more descriptive around major in-game events (Mateo-Orcajada et al., 2023). However, existing datasets do not sufficiently capture these characteristics of broadcast language, which limits their ability to support interaction-aware contextual caption generation. To address this limitation, we construct a large-scale game video dataset based on LoL, one of the most popular games worldwide. League of Legends continuously provides publicly accessible official broadcast videos, enabling large-scale data collection. Major in-game events are visualized on the screen in real time, which facilitates precise event-driven analysis. The game is a multiplayer online battle arena in which two teams of five players compete to destroy the opponent’s core structure, known as the Nexus. Throughout a match, interaction events such as champion kills and neutral object captures occur repeatedly. These events strongly influence the flow of the game and match outcomes, and they appear as clear visual and contextual cues in broadcast videos.

The dataset consists of official broadcast videos from 1,170 matches across five regular seasons of the League of Legends Champions Korea, spanning from the 2022 Summer Split to the 2024 Summer Split. All videos were collected at 1080p resolution and 59.94 frames per second (fps), with a total duration of approximately 696 hours. Broadcast audio was transcribed using the Whisper model (Radford et al., 2023), followed by post-processing to remove non-verbal utterances. Game commentary frequently includes proper nouns such as team names, player names, and champion

names, which vary across matches and may introduce bias during training. To mitigate this issue, these entities were anonymized using the tokens [TEAM], [PLAYER], and [CHAMP]. These tokens are among the most frequently occurring terms in the commentary captions, as illustrated in Figure 2f. The final dataset contains 31.16k refined commentary captions.

Statistical analysis shows that match durations are primarily distributed between 25 and 40 minutes, as shown in Fig. 2a. Commentary utterance frequency peaks between 20 and 35 minutes, when major team fights occur most frequently. Interaction event logs are therefore concentrated within the same time interval, as illustrated in Figure 2b. Most commentary sentences generated for Champion_kill and Object_kill events contain between 15 and 30 words, as illustrated in Figure 2c and 2d. In addition, distinct vocabularies are associated with different event types, indicating that the dataset consistently captures event-specific linguistic patterns, as summarized in figure 2g and 2h.

4. Methodology

Our goal is to enable short-form content creation by selecting meaningful highlight segments from full gameplay videos and generating context-aware captions for the selected segments. To this end, we propose a model that explicitly represents interactions formed by collections of game event logs and generates captions for the corresponding interaction segments. As illustrated in Figure 1, the proposed model consists of two main modules. The Interaction Module first detects individual game events represented as logs from the entire gameplay video and identifies interaction segments based on their temporal density. The Caption Generation Module then produces natural language descriptions

conditioned on the selected interaction segments, capturing the contextual and strategic dynamics of gameplay.

4.1. In-Game Event Log Detection

Let $\mathcal{V} = \{V_1, V_2, \dots, V_M\}$ denote a set of match videos, where each video V_i consists of a sequence of k frames, represented as $V_i = [v_i^1, v_i^2, \dots, v_i^k]$. From each video, events are identified by analyzing icon-based visual elements rendered in the game interface and are used to construct a temporally ordered sequence of interaction events for further processing.

Following prior work on minimap-based object detection (Kim et al., 2025), we formulate event log icon recognition as an object detection task. We employ YOLOv5 extra-large (YOLOv5x) as the detection backbone (Jocher, 2020). This formulation is robust to varying HUD layouts, where small icons in the event log and minimap regions make background areas prone to false detections. In contrast to classification-based approaches using CNN architectures such as ResNet (Hong & Lee, 2022), the detection-based formulation explicitly models overlapping icon instances and emphasizes icon-background separation through objectness learning. Combined with these design choices, the proposed formulation reduces false positives and enables reliable event log recognition.

Object Detection. Synthetic minimap images are generated from official game assets using a procedure comparable to prior minimap-based detection work. Unlike the prior work that focuses only on champion icons, the synthetic data is extended to include both champion and neutral object icons, covering the full set of interaction events of interest. As a result, the training data naturally contains small icon instances, partial overlaps, and globally applied visual noise that induces ambiguity between icons and background regions. To account for these characteristics, the detection model is trained with an objectness-focused objective, in which a relatively large weight is assigned to the objectness loss ($\lambda_{\text{obj}} = 5.0$), while bounding box regression and classification follow standard loss settings. This training strategy encourages reliable foreground-background discrimination and enables effective generalization from synthetic minimap inputs to real broadcast video frames. Interaction event logs are extracted from broadcast gameplay videos using the trained detector with class reduction and minimap-based pop-up detection, as detailed in Appendix A.

4.2. Salient Interaction Segment Selection

To identify temporally salient interaction segments from a game video, interaction events detected from the broadcast are represented as a continuous temporal signal using Gaussian kernel density estimation (KDE) (Parzen, 1962b; Davis et al., 2011a). KDE is adopted as a standard smooth-

ing technique to capture local concentrations of interaction events along the video timeline. Let $\{\ell_1^i, \dots, \ell_{N_i}^i\}$ denote the timestamps of interaction events extracted from video V_i . The interaction density is then defined as

$$\hat{f}^i(x) = \frac{1}{N_i h} \sum_{k=1}^{N_i} K\left(\frac{x - \ell_k^i}{h}\right), \quad (1)$$

where $K(\cdot)$ denotes a Gaussian kernel, defined as $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$, and h is the bandwidth parameter.

The bandwidth h is selected by scaling a robust global statistic of inter-event intervals to reflect typical temporal spacing of interactions. Salient interaction segments $\{S_j^i\}_{j=1}^{n_i}$ are then identified using a density-based threshold on $\hat{f}^i(x)$ and a minimum interaction constraint, with implementation details provided in Appendix B.

4.3. Interaction-aware Caption Generation

Video-Text Alignment Learning. Given an event occurring at time t , a clip $V = \{f_1, f_2, \dots, f_L\}$ is formed by uniformly sampling L frames at 1 fps from the preceding temporal window $[t - \Delta, t]$, where Δ denotes the duration of the temporal window used for event encoding. Each frame f_i is independently processed by a pre-trained visual backbone $\mathcal{F}(\cdot)$, producing a frame-level feature $\mathbf{h}_i = \mathcal{F}(f_i) \in \mathbb{R}^{d_v}$. The frame-level features are aggregated using temporal attention pooling to obtain an event-level visual representation $\mathbf{v} = \sum_{i=1}^L \alpha_i \mathbf{h}_i$. The pooled feature $\mathbf{v}$ is projected into a shared embedding space using a two-layer MLP $g_v(\cdot)$ and L2-normalized, yielding the final event embedding $\mathbf{z}_v \in \mathbb{R}^d$ with $d = 512$. Each interaction segment is represented as a set of such event embeddings.

Captions are encoded using a BERT-based text encoder (Devlin et al., 2019). Given a caption C , we extract the [CLS] token representation from the final transformer layer, denoted as $\mathbf{t} \in \mathbb{R}^{768}$, where 768 is the hidden dimension of BERT-base. This representation is projected into the shared embedding space through a learnable projection head $g_t(\cdot)$ and L2-normalized to obtain the text embedding $\mathbf{z}_t \in \mathbb{R}^d$.

To align video and caption representations, we use a symmetric contrastive loss. Given a mini-batch of N paired video clips and captions $\{(V_i, C_i)\}_{i=1}^N$, we compute the pairwise similarity between video and text embeddings as $s_{ij} = \tau \mathbf{z}_{v_i}^\top \mathbf{z}_{t_j}$, where τ denotes a learnable temperature parameter. The contrastive objective encourages matched video-caption pairs to have higher similarity than mismatched pairs within the batch. Specifically, we optimize bidirectional retrieval losses from video to text and from text to video, defined as

$$\mathcal{L}_{v \rightarrow t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ij})}, \quad (2)$$

and

$$\mathcal{L}_{t \rightarrow v} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ji})}. \quad (3)$$

The final contrastive loss is obtained by averaging the two directions, $\mathcal{L}_{\text{cont}} = \frac{1}{2}(\mathcal{L}_{v \rightarrow t} + \mathcal{L}_{t \rightarrow v})$.

Caption Generation. The embedding $\mathbf{z}_v \in \mathbb{R}^d$, obtained through video-text alignment learning, encodes event-relevant visual information. To condition caption generation on visual context, $\mathbf{z}_v$ is projected into the hidden space of the text model, yielding a visual conditioning vector $\mathbf{e}_v$.

We evaluate three architectures for caption generation: LSTM (Hochreiter & Schmidhuber, 1997), mLSTM (Beck et al., 2024), and a GPT-2-(Radford et al., 2019) based transformer decoder. To provide explicit event semantics, an event type token e is prepended to the caption sequence. Specifically, the input takes the form $[\mathbf{e}_v; \mathbf{E}_e; \mathbf{E}_{\text{sos}}; \mathbf{E}_{w_1}; \dots; \mathbf{E}_{w_L}]$, where $\mathbf{e}_v$ provides visual context and $\mathbf{E}_e$ encodes the event type. For recurrent architectures, including LSTM and mLSTM, the visual embedding $\mathbf{e}_v$ conditions the initial recurrent state, and captions are generated autoregressively. For the transformer-based model, $\mathbf{e}_v$ is prepended as a visual prefix token, enabling self-attention to visual context across all layers. All models are trained with a standard autoregressive cross-entropy objective, conditioning each prediction on the visual embedding, the event type, and previously generated tokens.

5. Experiments

5.1. Implementation Details

Loss Configuration. To account for the visual ambiguity between foreground objects and background patterns in the minimap-based detection setting, we assign a substantially larger weight to the objectness loss. Specifically, the objectness loss weight is set to 5.0, compared to 0.05 for the bounding box regression loss and 0.5 for the classification loss. This design encourages the model to prioritize accurate object presence prediction. We further adopt relaxed target assignment criteria, with an anchor overlap threshold of 0.20 and an anchor size tolerance of 4.0, allowing a broader set of candidate locations to be treated as positive samples during training.

Backbone and Pretraining. To extract visual representa-

Table 3. Event detection performance under different temporal tolerances (in seconds)

Temporal Tolerance	Precision	Recall	F1
1	94.97	92.80	93.87
3	98.01	95.68	96.83
5	98.60	96.25	97.41

Table 4. Event detection performance by season and event type

Season	Metric	Overall	Champion.kill	Object.kill		
				Overall	Dragon	Baron
2022 Summer	TP	4845	3606	1239	943	296
	FP	470	463	7	5	2
	FN	510	496	14	12	2
	Precision	91.16	88.62	99.44	99.47	99.33
	Recall	90.48	87.91	98.88	98.74	99.33
2023 Spring	F1	90.82	88.26	99.16	99.11	99.33
	TP	5699	4495	1204	931	273
	FP	263	221	42	29	13
	FN	232	205	27	17	10
	Precision	95.59	95.31	96.63	96.98	95.45
2023 Summer	Recall	96.09	95.64	97.81	98.21	96.47
	F1	95.84	95.48	97.21	97.59	95.96
	TP	5573	4512	1061	800	261
	FP	544	458	86	60	26
	FN	503	402	101	74	27
2024 Spring	Precision	91.11	90.78	92.50	93.02	90.94
	Recall	91.72	91.82	91.31	91.53	90.63
	F1	91.41	91.30	91.90	92.27	90.78
	TP	5887	4805	1082	879	203
	FP	40	27	13	13	0
2024 Summer	FN	193	164	29	22	7
	Precision	99.33	99.33	99.33	99.33	99.33
	Recall	96.83	96.70	97.39	97.56	96.67
	F1	98.06	97.99	98.35	98.43	97.98
	TP	5047	4282	765	765	161
Overall	FP	115	108	7	7	1
	FN	662	584	78	78	33
	Precision	97.77	97.54	99.09	99.09	99.38
	Recall	88.40	88.00	90.75	90.75	82.99
	F1	92.85	92.52	94.74	94.74	90.45
Overall	TP	27051	21700	5351	4318	1194
	FP	1432	1277	155	114	42
	FN	2100	1851	249	203	79
	Precision	94.97	94.44	97.18	97.43	96.60
	Recall	92.80	92.14	95.55	95.51	93.79
Overall	F1	93.87	93.28	96.36	96.46	95.18

tions, we employ three visual backbones in our experiments: ResNet-152 (He et al., 2016), ViT-B/16 with relative position bias (RelPos), and ViT-L/16 (Wightman, 2019; Dosovitskiy et al., 2021; Liu et al., 2022). All backbones are initialized with ImageNet-pretrained weights and subsequently adapted to the interaction-centric video domain via event-level video-text contrastive learning.

Frame Sampling. For each event, we uniformly sample $T = 20$ frames at 1 fps from a fixed temporal window of 20 seconds preceding the event timestamp. Sampling is restricted to the pre-event interval, as post-event frames predominantly contain replay footage and provide limited contextual information for modeling the event.

Data Preprocessing. To focus on the gameplay region while suppressing irrelevant interface elements, each sampled frame is center-cropped to 768×768 pixels, with the crop center aligned at 42% of the frame height. The cropped frames are then resized to 224×224 and normalized using ImageNet statistics before being fed into the visual encoder.

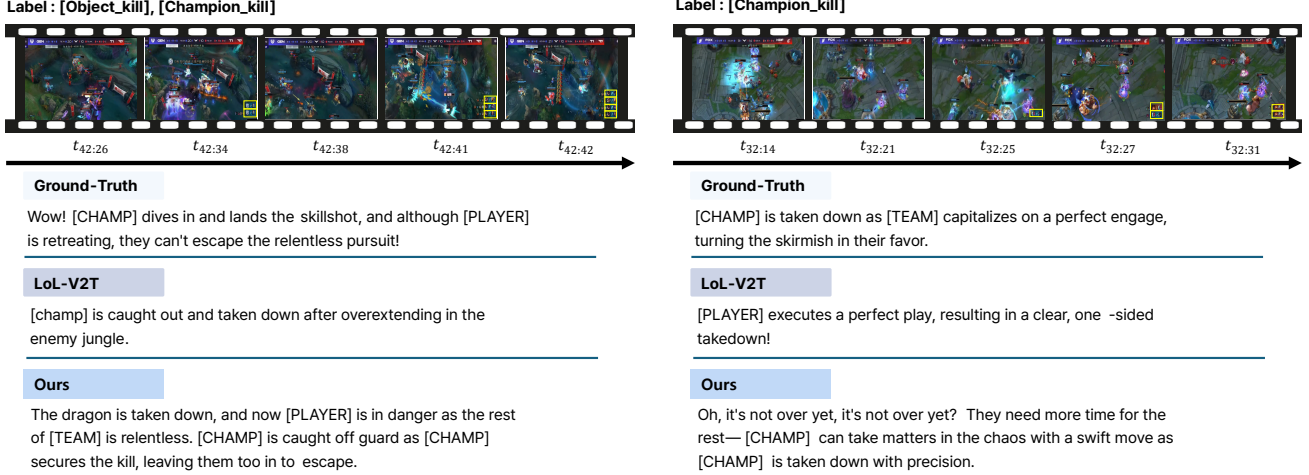


Figure 3. Qualitative results of interaction-aware caption generation

5.2. Highlight Selection Results

5.2.1. EVENT LOG DETECTION

Event log detection performance is evaluated using precision, recall, and F1-score. In game broadcasts, event logs typically remain visible on the screen for several seconds after their occurrence, with a duration of up to five seconds. To analyze temporal sensitivity, we evaluate event detection accuracy under temporal tolerances of 1, 3, and 5 seconds, where a prediction is considered correct if it falls within the corresponding window around the ground-truth timestamp. The proposed method achieves high detection accuracy, as summarized in Table 3. Even under a strict temporal tolerance of 1 second, indicating precise temporal localization. Detection performance further improves as the tolerance increases, with the highest scores observed at a tolerance of 5 seconds. However, a temporal deviation beyond 1 second often corresponds to a gameplay state in which the original event has already concluded. Therefore, to ensure accurate event localization and to avoid crediting detections after the event has effectively ended, we adopt a strict temporal tolerance of 1 second for all subsequent experiments.

Under a strict temporal tolerance of 1 second, detailed results are reported in Table 4. The Champion_kill event achieves a score of 93.28%, while the Object_kill event reaches 96.36%. A more fine-grained analysis shows that Dragon and Baron events attain scores of 96.46% and 95.18%, respectively. These results indicate consistent detection performance across different event types.

5.2.2. SALIENT INTERACTION SEGMENT SELECTION

Results of salient segment selection under nonlinearly distributed interactions are illustrated in Figure 4. Interaction density over time is estimated using KDE, with the x-axis denoting video time (in seconds) and the y-axis representing the normalized interaction density. Regions exceeding

a mean-based threshold are highlighted as selected salient segments. Panel (a) presents examples with densely clustered interactions, where pronounced density peaks result in coherent KIS selection. In contrast, panel (b) shows sparse interaction cases, reflecting substantial variability in interaction distributions across videos. In such scenarios, combining density thresholding with an interaction-count constraint effectively suppresses isolated interactions, leading to more reliable interaction segment selection. Overall, these results demonstrate stable highlight segment identification across diverse interaction patterns.

5.3. Caption Generation Results

5.3.1. QUANTITATIVE EVALUATION

We evaluate caption generation performance using standard language generation metrics, including BLEU, METEOR, ROUGE-L, and CIDEr. Performance varies noticeably depending on both the choice of visual encoder and text decoder, as summarized in Table 5. Across all visual backbones, LSTM-based decoders consistently achieve the strongest performance across most evaluation metrics, indicating their effectiveness in modeling temporally structured interaction sequences. GPT-2-based decoders achieve competitive results, but generally underperform LSTM-based models in the interaction-centric captioning setting. This trend suggests that recurrent architectures, which explicitly maintain a temporal state, are well suited for capturing short-range dependencies and ordered event progression commonly observed in gameplay commentary. Among visual encoders, ViT-B/16 with relative position bias yields the strongest overall performance, suggesting that explicit positional modeling is beneficial for capturing interaction-level semantics in long-form gameplay videos.

Compared to the baseline, the proposed method shows consistent improvements across most evaluation metrics, as

shown in Table 6. In particular, the ViT-B/16 + RelPos + LSTM configuration achieves higher BLEU, METEOR, and ROUGE scores, demonstrating the effectiveness of interaction-aware highlight selection and context-aware caption generation. While the MART-based baseline attains a higher CIDEr score, our method performs better on METEOR and ROUGE, indicating stronger semantic adequacy and contextual coherence.

5.3.2. ABLATION STUDY

An ablation study is conducted to analyze the effects of pretraining, prefix tuning, and teacher forcing strategies on caption generation performance. Applying both pretraining and prefix tuning yields the best overall performance across all evaluation metrics, indicating their complementary effects, as shown in Table 9. In particular, incorporating event tokens provides explicit semantic cues about underlying game events, allowing the model to generate captions that better align with the intended gameplay context. In addition, the impact of different teacher forcing (TF) ratios is evaluated. A TF ratio of 0.7 achieves the best overall performance, while higher or lower ratios lead to slight performance degradation, suggesting that moderate teacher forcing is most effective for stable training, as shown in Table 10.

5.3.3. QUALITATIVE ANALYSIS

Qualitative comparisons between the baseline and our method are presented over interaction segments, as shown in Fig. 4. In the left example, the interaction segment includes both `Object_kill` and `Champion_kill` events, where the ground-truth commentary relates the kill to the surrounding objective-focused context. While LoL-V2T generates a generic kill-centric description that overlooks this interaction, our method explicitly connects the takedown to the objective-related flow, resulting in a more context-faithful caption. In the right example, multiple temporally adjacent `Champion_kill` events are grouped into a single interaction segment. Our method produces a concise caption that preserves the causal intent of the play, whereas the baseline tends to output a templated description with weaker grounding to the event sequence. Overall, these examples demonstrate that interaction-aware conditioning improves contextual coherence, which is further supported by a user study evaluating the practical usability of extracted highlights and generated captions, as reported in Table 8.

5.4. Discussion

Data Applicability. Our model operates directly on publicly available broadcast gameplay videos by detecting on-screen event logs, without relying on game application programming interfaces (APIs) or internal telemetry. As a result, the

Table 5. Encoder-decoder comparison. Caption generation performance for different combinations of visual encoders and text decoders. We report BLEU, METEOR, ROUGE, and CIDEr scores, and bold the best results.

Visual Encoder	Text Decoder	Commentary Metrics				
		B@3	B@4	M	R	C
ResNet-152	LSTM	3.22	1.52	15.85	15.02	7.53
	GPT2	2.84	1.41	15.26	13.01	6.19
	xLSTM	2.30	1.13	15.64	14.07	3.69
ViT-B/16 + RelPos	LSTM	3.17	1.54	16.05	15.31	7.14
	GPT2	2.92	1.40	15.14	13.01	5.97
	xLSTM	1.96	0.91	16.34	13.88	3.02
ViT-L/16	LSTM	2.73	1.27	15.51	14.36	6.62
	GPT2	2.51	1.20	14.53	12.39	5.21
	xLSTM	1.61	0.63	15.20	13.47	2.37

Table 6. Caption generation performance. The LoL-v2t results are reproduced following the official settings of the original paper.

Method	Visual Encoder	Text Decoder	Captioning Metrics				
			B@3	B@4	M	R	C
LoL-V2T (Tanaka & Simo-Serra, 2021)	ResNet	LSTM	2.29	1.30	13.91	12.29	7.78
		MART	2.17	1.25	12.26	12.66	8.34
Ours	ViT-B/16 + RelPos	LSTM	3.17	1.54	16.05	15.31	7.14
	ResNet	LSTM	3.22	1.52	15.85	15.02	7.53
	ResNet	GPT-2	2.84	1.41	15.26	13.01	6.19

proposed approach can be applied to games that do not provide official APIs or structured gameplay data. Although our experiments focus on League of Legends, the model generalizes to other esports titles that employ explicit HUD-based event visualizations.

Limitations. The proposed method relies on the visual availability of on-screen event logs and may be sensitive to variations in video quality or broadcast layouts. In addition, interactions that are not explicitly reflected by event logs, such as long-term strategic dynamics, are not fully captured by the current model. While our minimap-based detection design improves robustness to some extent, modeling richer and longer-range interactions remains an important direction for future work.

Challenges and Future Directions. Future work includes incorporating additional modalities, such as audio cues, modeling longer temporal dependencies, and extending the model to other game domains.

6. Conclusion

In this work, we propose an interaction-aware model for highlight selection and caption generation that captures the contextual structure of game videos beyond individual events. By modeling interaction units and applying KDE-based segment selection, the proposed model effectively represents temporal dependencies and strategic flows. Experiments on large-scale game videos show that our approach outperforms the baseline and enables practical short-form content generation from real gameplay videos.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Inference Procedure for Event Logs

In this section, we detail the inference procedure for detecting interaction event logs from broadcast gameplay videos. Implementation-level descriptions of the minimap mapping process and the interaction pop-up detection logic are provided for completeness. These procedures correspond to Figure 4 and are summarized in Algorithms 1 and 2.

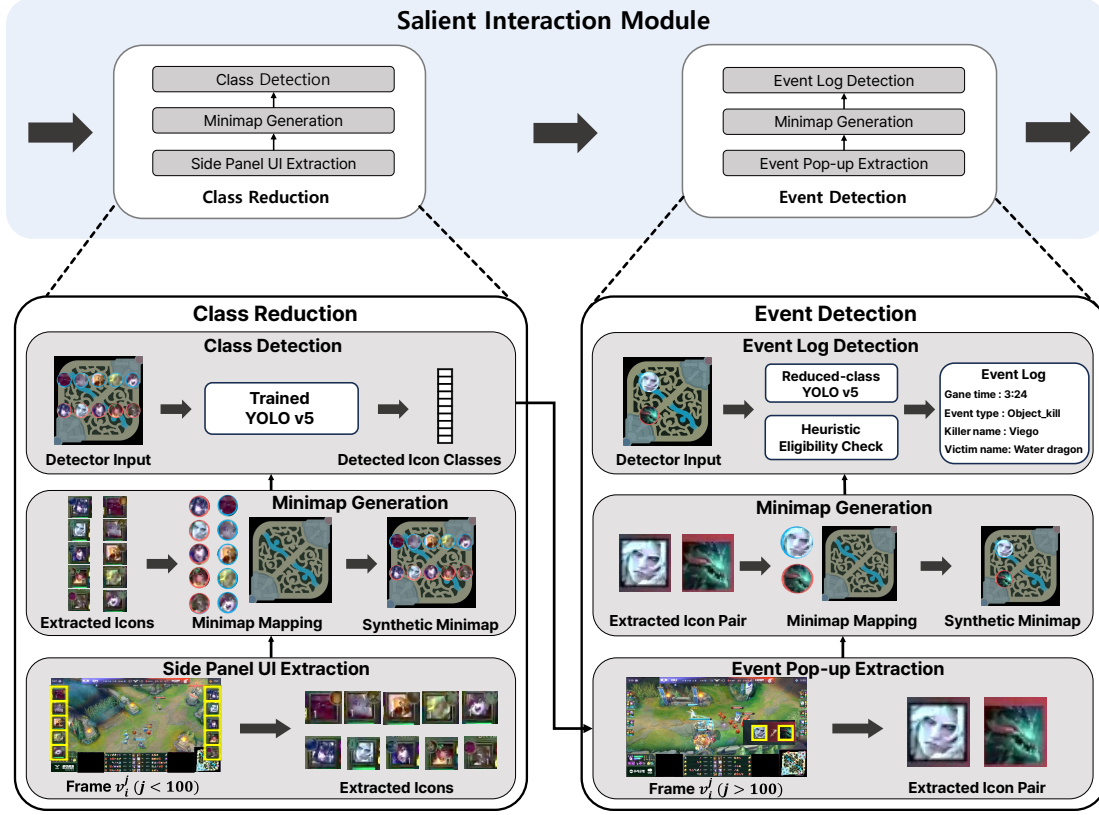


Figure 4. Overview of Salient Interaction Module

Algorithm 1 Minimap mapping

- 1: **Input:** Icon image r_i
- 2: **Output:** Synthetic minimap M'
- 3: $M \leftarrow$ load base minimap from game assets
- 4: $r_i \leftarrow$ resize r_i to fixed size and apply circular mask
- 5: $(x, y) \leftarrow$ compute coordinates from team/slot
- 6: Overlay r_i onto M at (x, y) with team-colored outline
- 7: **return** M'

Class Reduction. The detection model is trained to recognize a comprehensive set of icon classes, including 169 champion icons and 8 neutral object icons, whereas each match video V_i contains only 10 champions together with neutral objects. This mismatch motivates a Class Reduction step, which constrains the detection space at inference time. To this end, the champion classes actually appearing in the video are identified before event log detection. During the first 100 frames $[v_i^1, \dots, v_i^{100}]$, champion icons are cropped from the side panel UI and mapped onto the minimap using Algorithm 1. The resulting synthetic minimap inputs are processed by the trained detector, and predictions are aggregated across frames to obtain a consistent set of champion classes. This reduced class set is then used throughout subsequent inference, effectively eliminating absent icon classes and improving robustness against spurious detections.

Algorithm 2 Interaction pop-up detection

Input:

v_i^t : input frame
 M_{YOLO} : detection model
 $\mathcal{M}$: timestamp memory, $\mathcal{M}[c] = -\infty$ for all classes c
 θ : confidence threshold ($\theta = 0.9$)

Output:

I_t : detected interaction log at frame t

```

1:  $I_t \leftarrow \emptyset$  // Eligible pair
2: if  $t \bmod 30 \neq 0$  then
3:   return  $I_t$ 
4: end if
5:  $P \leftarrow$  crop popup icon pairs from  $v_i^t$ 
6:  $V \leftarrow \emptyset$  // Valid pair
7: for  $r = 0$  to 3 do
8:   if  $r \geq 2$  and  $P[r-1] \notin V$  then
9:     break
10:  end if
11:   $(k, v) \leftarrow P[r]$  // ( $k$ : killer,  $v$ : victim)
12:   $M' \leftarrow$  Minimap mapping( $k, v$ ) // Algorithm 1
13:   $D \leftarrow M_{YOLO}.detect(M')$ 
14:   $d_k, d_v \leftarrow$  extract detections for  $(k, v)$  from  $D$ 
15:  if  $d_k.conf > \theta$  and  $d_v.conf > \theta$  then
16:     $V \leftarrow V \cup \{(k, v)\}$ 
17:    if  $d_v.class \neq c_{object}$  then
18:       $h_v \leftarrow$  health_point of  $d_v$ 
19:       $\Delta t \leftarrow t - \mathcal{M}[d_v.class]$ 
20:      if  $h_v < 0.05$  and  $\Delta t > 10s$  then
21:         $I_t \leftarrow I_t \cup \{(d_k.class, d_v.class)\}$ 
22:         $\mathcal{M}[d_v.class] \leftarrow t$ 
23:      end if
24:    else
25:       $\Delta t \leftarrow t - \mathcal{M}[d_v.class]$ 
26:      if  $\Delta t > 30s$  then
27:         $I_t \leftarrow I_t \cup \{(d_k.class, d_v.class)\}$ 
28:         $\mathcal{M}[d_v.class] \leftarrow t$ 
29:      end if
30:    end if
31:  end if
32: end for
33: return  $I_t$ 
    
```

Event Detection. Given the reduced class set, the Event Detection step extracts interaction events directly from broadcast frames. In each frame v_i^j of video V_i , interaction pop-up logs appear at fixed locations in the bottom-right corner as ordered (killer, victim) icon pairs. Starting from frame 100, frames are sampled at 2 fps, and the cropped icon pairs are projected onto the minimap using Algorithm 1 prior to detection. To suppress false detections, lower-row pairs are evaluated only if the preceding row is valid, and repeated detections caused by the temporal persistence of pop-up logs are filtered using class-dependent health constraints and a timestamp memory. Only detections satisfying these spatial and temporal conditions are retained as valid interaction events. The overall procedure is illustrated in Figure 4 and summarized in Algorithm 2.

Table 7. Mean performance across different KDE bandwidths.

h	Precision	Recall	F1
1.0	0.9941	0.1089	0.1945
3.2	0.9480	0.2123	0.3433
5.0	0.9581	0.1682	0.2828
7.0	0.9209	0.2555	0.3962
10.0	0.9283	0.2425	0.3819
15.0	0.9581	0.1682	0.2828
20.0	0.9853	0.1323	0.2258

B. Details of Salient Interaction Segment Selection

Bandwidth selection. Let $\{\ell_k^i\}$ denote the timestamps of interaction logs in video V_i . We first compute the set of inter-event intervals $\{\ell_{k+1}^i - \ell_k^i\}$ for each video. From these intervals, the first quartile is computed and denoted as Q_1^i . To obtain a global representative statistic that is robust to outliers across videos, we define

$$Q_1^* = \text{median}(\{Q_1^i\}_{i=1}^M).$$

The kernel bandwidth h is then determined as

$$h = 0.8 \times Q_1^*.$$

This scaling factor produces a slightly narrower bandwidth, allowing the kernel density estimation to be more sensitive to temporally dense interaction patterns. In our experiments, Q_1^* was computed as 4.0 seconds, yielding $h = 3.2$ seconds.

Density thresholding. Given the kernel density estimate $\hat{f}^i(x)$ for video V_i , a density-based threshold θ^i is defined as

$$\theta^i = \alpha \cdot \text{mean}(\hat{f}^i(x))$$

where α is a constant factor. We set $\alpha = 5.0$ in all experiments. This choice reflects the strong imbalance between interaction and non-interaction periods in gameplay videos, ensuring that only temporally salient regions with substantially higher-than-average interaction density are selected.

Minimum interaction constraint. To suppress isolated or spurious detections, each candidate segment is required to contain at least two interaction logs. Segments containing only a single interaction are discarded, as they are unlikely to represent meaningful multi-agent interactions.

Segment selection. Only temporal intervals that satisfy both the density thresholding criterion and the minimum interaction constraint are retained. The resulting intervals are treated as key interaction segments and are used in subsequent highlight selection and caption generation stages. The KDE visualizations shown in Fig. 8 illustrate the final density estimation results obtained with the selected bandwidth and threshold, and directly correspond to the segments retained by this procedure.

Bandwidth sensitivity analysis (quantitative). To validate the appropriateness of the selected bandwidth, we evaluate multiple KDE bandwidths using a manually annotated highlight ground-truth set constructed from curated YouTube highlight videos of the same matches. For each match, highlight moments are identified from the edited highlight video, their in-game timestamps are manually recorded, and then aligned to the corresponding full broadcast videos in our dataset via a synchronization point between game time and video time. For each bandwidth, interaction segments are generated and compared against these ground-truth highlight intervals using temporal overlap-based precision, recall, and F1. As shown in Fig. 5 and Table 7, increasing the bandwidth produces smoother density estimates and broader segments, which generally improves recall by covering more of the ground-truth intervals, but may reduce precision due to the inclusion of temporally irrelevant regions. Conversely, smaller bandwidths yield more localized segments that better preserve precision but often fragment salient interactions, resulting in lower recall.

Although larger bandwidths (e.g., $h = 7.0$) can achieve higher F1 scores, we select $h = 3.2$ as a conservative operating point. This choice maintains high precision while providing competitive recall and F1, leading to temporally tighter interaction segments that better align with the goal of concise highlight localization. Overall, the results demonstrate that bandwidth

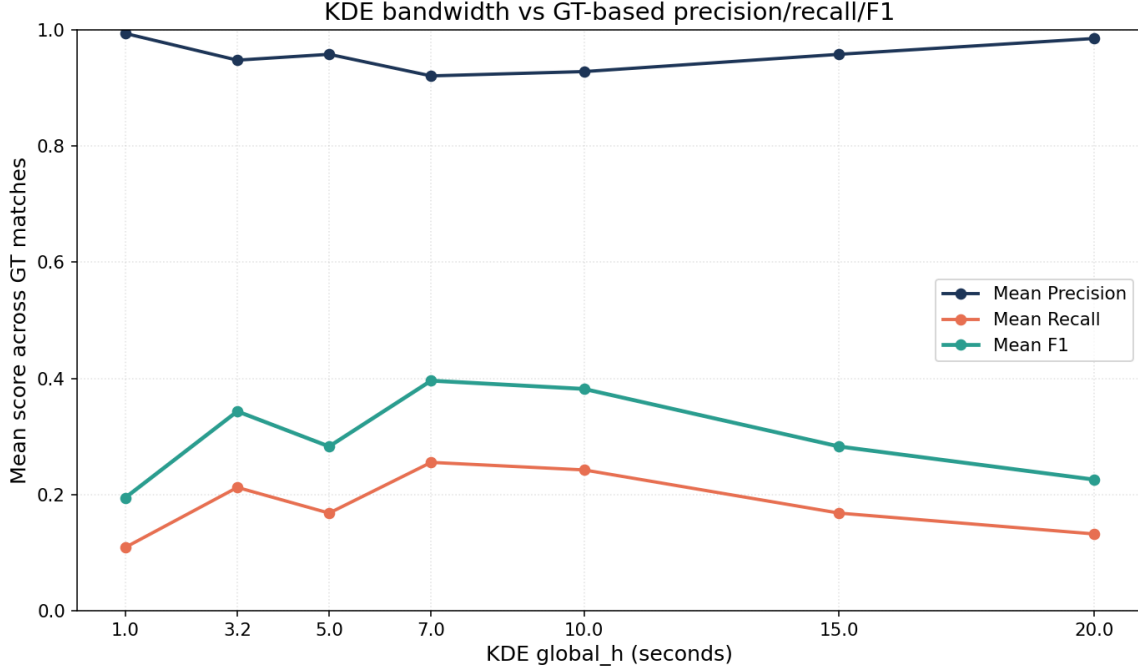


Figure 5. KDE bandwidth sensitivity analysis. As the bandwidth increases, recall improves while precision slightly decreases, illustrating a clear precision–recall tradeoff. Although $h = 7.0$ achieves the highest mean F1, we select $h = 3.2$ as a conservative operating point that preserves precision while maintaining competitive recall.

controls a clear precision–recall tradeoff, and we prioritize a precision-preserving setting with stable performance in the main experiments.

Bandwidth sensitivity analysis (qualitative). As shown in Fig. 7, the effect of bandwidth is clearly observable in the temporal structure of predicted segments relative to the ground-truth highlight intervals. Smaller bandwidths (e.g., $h = 1.0$) produce highly fragmented segments that fail to capture complete interactions, while larger bandwidths (e.g., $h = 15.0$ and $h = 20.0$) merge multiple events into overly broad intervals, often extending beyond the annotated highlight regions.

In contrast, the proposed bandwidth ($h = 3.2$) achieves a better balance, producing segments that are both temporally precise and well-aligned with the ground-truth intervals derived from curated highlight videos. This observation is consistent with the quantitative results, supporting our choice of $h = 3.2$ as a precision-preserving operating point.

Multi-method qualitative comparison. We compare KDE-based segment extraction with DBSCAN and sliding window baselines. As shown in Fig. ??, DBSCAN produces sparse detections and often fails to capture many ground-truth highlight intervals, resulting in low recall. Sliding window produces segments with better coverage, but lacks temporal adaptivity, leading to imprecise boundaries and reduced precision.

In contrast, the KDE-based method produces segments that are well-aligned with the ground-truth highlights in both location and duration. By adapting to the density of interaction events, KDE captures the temporal structure of salient interactions more effectively. This behavior is consistently observed across multiple matches, as further illustrated by the corresponding gameplay frames.

We conducted a user study with 20 participants during a public game exhibition. Participants evaluated automatically generated short-form gameplay videos together with their corresponding narrative captions, produced from gameplay footage recorded by non-expert players. All evaluated highlight segments and captions were generated using the full interaction-aware pipeline described in this paper.

Table 8. User study results on automatic short-form caption generation from real gameplay videos.

Evaluation Item	Average Score / Max	Percentage
Need for enhanced immersion through automatic narration	4.25 / 5	85%
Naturalness and enjoyment of generated narratives	4.55 / 5	90%
Distinctiveness and characterization of narrative content	4.35 / 5	87%
Appropriateness of narration timing during gameplay	4.41 / 5	88%
Contribution to overall gameplay experience	4.28 / 5	86%
Average gameplay duration	Over 54 minutes	
Qualitative Feedback Summary		
Participants emphasized that narrative generation should primarily focus on key interaction events.		
Several noted that narration must be carefully timed to avoid interrupting gameplay immersion.		
Support for multiple game genres was suggested as a direction for broader applicability.		
Some participants reported that poorly timed narration could negatively affect player experience.		
Overall, users perceived the system as beneficial for enhancing engagement and narrative completeness.		
Concise and summarized narration was also recommended to further improve usability.		

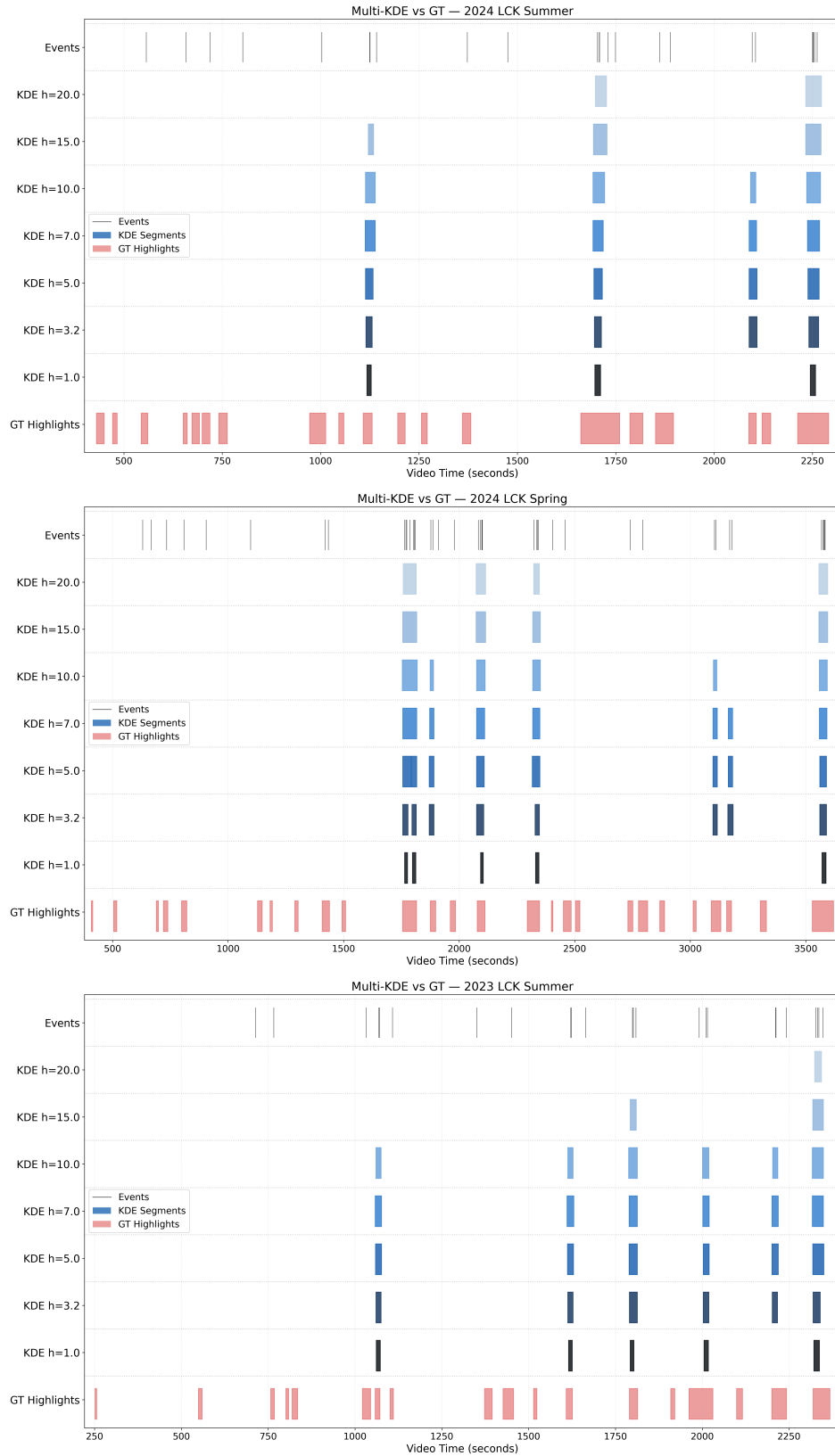


Figure 6. Qualitative comparison of KDE-based interaction segments across different bandwidths. Black lines indicate interaction events, blue rectangles denote predicted segments, and red rectangles represent ground-truth highlights. Smaller bandwidths produce fragmented segments, while larger bandwidths generate overly broad segments. The selected bandwidth ($h = 3.2$) yields temporally concise segments that align well with ground-truth intervals.

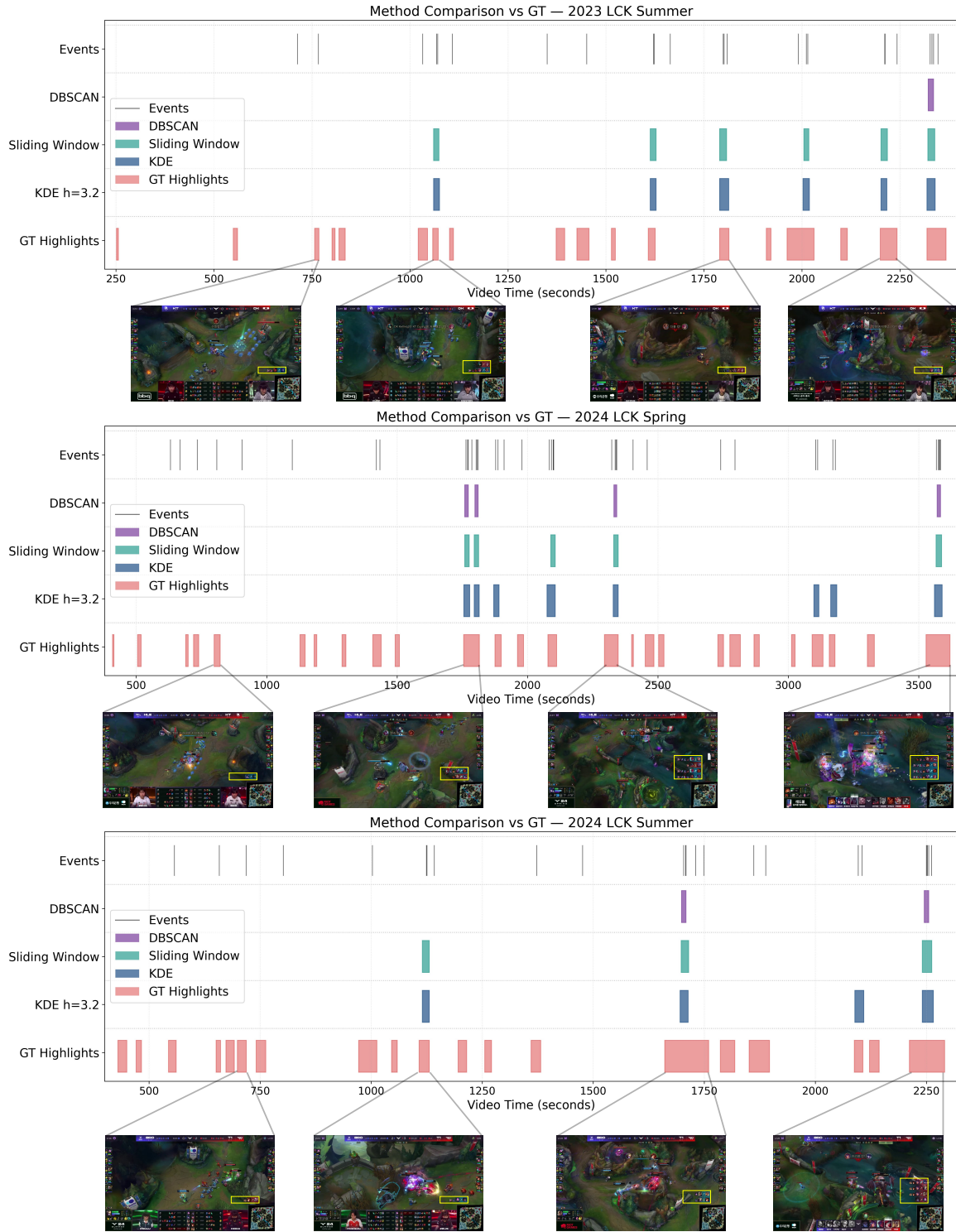
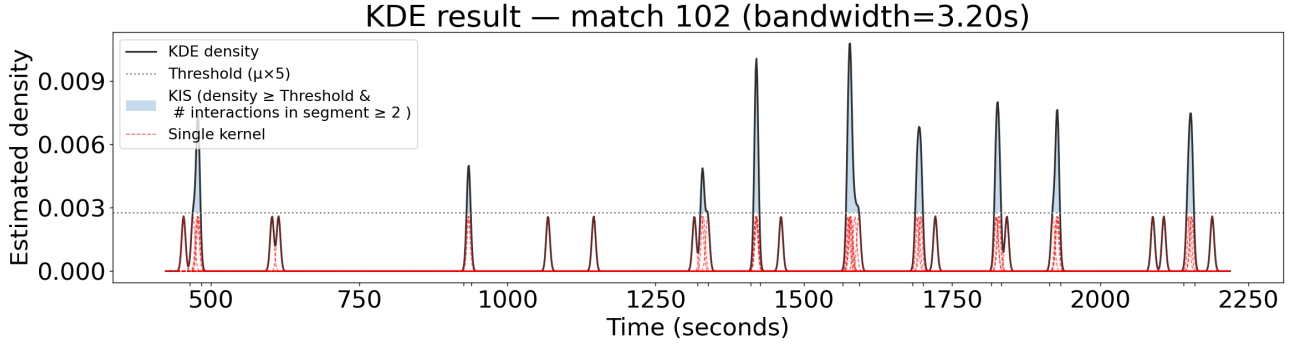
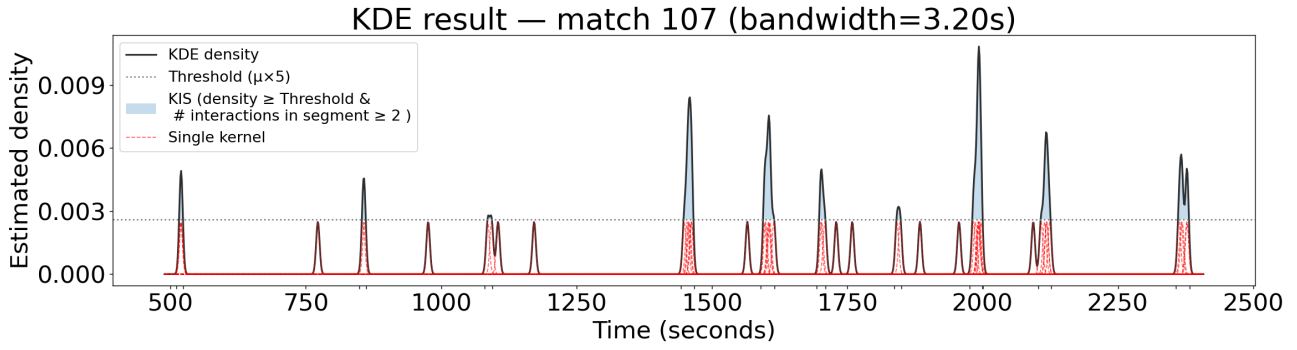


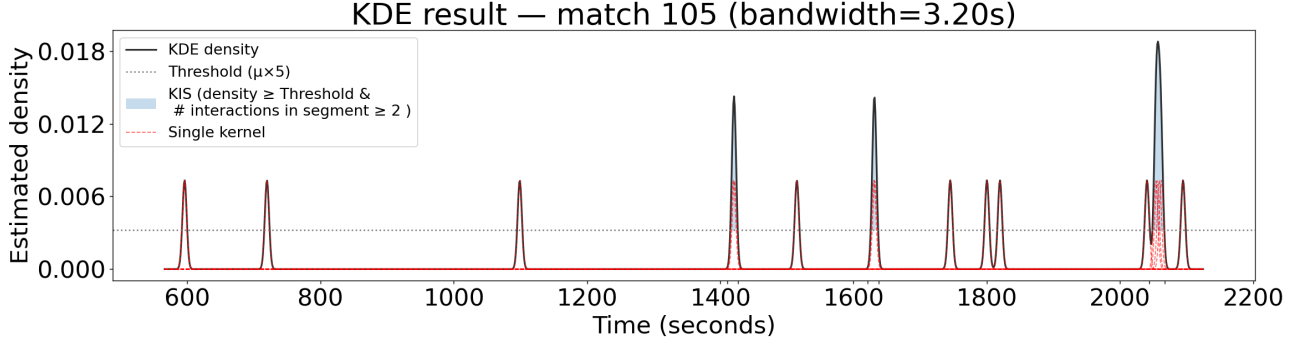
Figure 7. Qualitative comparison of segment extraction methods across multiple matches. Black lines denote interaction events, while colored rectangles indicate predicted segments (DBSCAN: purple, sliding window: green, KDE: blue) and ground-truth highlights (red), where ground-truth intervals are derived from curated highlight videos and aligned to the broadcast videos. DBSCAN produces sparse detections, and sliding window yields imprecise temporal boundaries, whereas the KDE-based method ($h = 3.2$) generates segments that best align with ground-truth highlights in both location and duration.



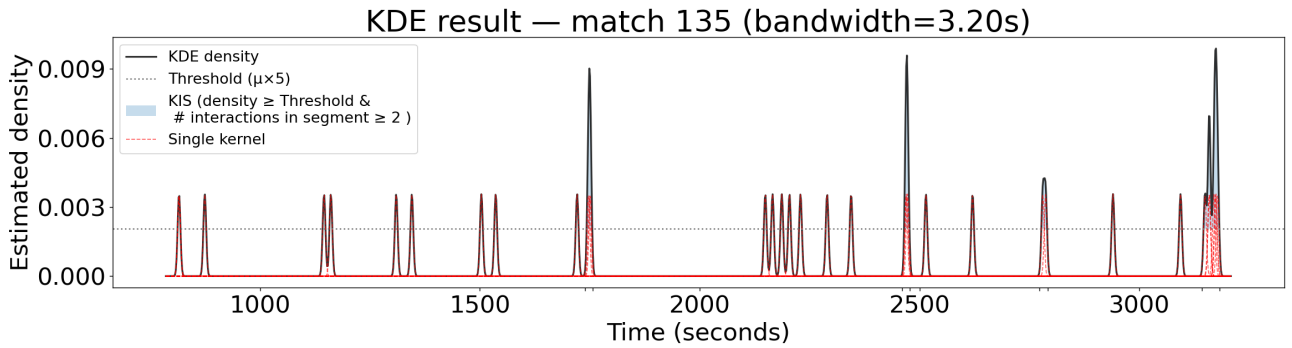
(a) Dense interaction video (example 1)



(b) Dense interaction video (example 2)



(c) Sparse interaction video (example 1)



(d) Sparse interaction video (example 2)

Figure 8. KDE examples for interaction distributions in videos.

C. Ablation Study on the Caption Generation Module

Table 9. Ablation study on pretraining and prefix tuning strategies

Pretrain	Prefix	B@3	B@4	M	R	C
×	×	2.56	1.05	13.40	12.73	5.11
✓	×	2.42	1.10	13.91	14.18	6.59
×	✓	2.50	1.03	16.06	14.43	5.33
✓	✓	3.17	1.54	16.05	15.31	7.14

Table 10. Ablation study on teacher forcing (TF) ratios.

TF	B@3	B@4	M	R	C
1.0	2.94	1.41	15.54	14.95	7.30
0.7	3.17	1.54	16.05	15.31	7.14
0.5	2.98	1.39	15.87	14.75	6.35
0.3	2.90	1.31	15.11	14.73	6.33

Table 11. Result of per-class detection performance

Name	precision	recall	f1_score	Name	precision	recall	f1_score	Name	precision	recall	f1_score	Name	precision	recall	f1_score
Aatrox	96.67	94.69	95.67	Ahri	88.96	89.68	89.32	Akali	90.48	89.24	89.85	Akshan	100.00	100.00	100.00
Amumu	75.81	79.66	77.69	Annie	95.08	95.08	95.08	Aphelios	89.67	90.57	90.12	Ashe	96.25	92.93	94.56
Aurora	100.00	94.37	97.10	Azir	93.81	91.06	92.41	Bard	97.56	93.02	95.24	Belveth	94.44	94.44	94.44
Brand	93.90	88.51	91.12	Braum	93.90	93.90	93.90	Caitlyn	98.52	98.52	98.52	Camille	96.30	87.39	91.63
Chogath	85.71	75.00	80.00	Corki	95.49	85.28	90.10	Darius	97.14	97.14	97.14	Diana	93.75	93.75	93.75
Elise	96.99	98.47	97.73	Ezreal	97.03	95.03	96.02	FiddleSticks	100.00	100.00	100.00	Flora	96.60	96.60	96.60
Gangplank	87.86	86.75	87.30	Garen	100.00	100.00	100.00	Gnar	97.47	97.06	97.26	Gragas	87.50	87.04	87.27
Gwen	91.60	89.78	90.68	Hecarim	60.00	85.71	70.59	Heimerdinger	99.23	99.23	99.23	Hwei	96.93	91.33	94.05
Irelia	88.89	93.33	91.06	Ivern	94.05	91.86	92.94	JarvanIV	96.67	96.67	96.67	Jax	96.48	92.99	94.70
Jhin	92.58	86.18	89.26	Jinx	87.57	85.16	86.35	KSante	94.22	89.93	92.03	Kaisa	94.34	93.46	93.90
Karma	96.23	95.20	95.71	Karthus	82.76	80.00	81.36	Kassadin	96.55	97.39	96.97	Kayle	96.30	96.30	96.30
Khazix	90.60	87.60	89.08	Kindred	86.89	88.33	87.60	KogMaw	89.47	89.47	89.47	Leblanc	89.95	84.22	86.99
Leona	96.92	94.19	95.54	Lillia	96.85	84.83	90.44	Lissandra	94.68	93.68	94.18	Lucian	93.80	93.10	93.45
Lux	94.19	98.78	96.43	Malphite	92.19	92.19	92.19	Maokai	94.47	93.86	94.17	Milio	99.31	99.31	99.31
MonkeyKing	91.38	92.10	91.74	Mordekaiser	86.36	84.44	85.39	Morgana	100.00	97.44	98.70	Naafiri	100.00	100.00	100.00
Nasus	100.00	100.00	100.00	Nautilus	94.73	92.86	93.78	Neeko	95.19	94.29	94.74	Nidalee	93.98	75.30	83.61
Nocturne	94.72	94.72	94.72	Olaf	96.83	91.04	93.85	Orianna	99.55	97.79	98.66	Ornn	94.06	93.73	93.89
Poppy	89.02	86.99	87.99	Pyke	93.55	96.67	95.08	Qiyana	33.33	33.33	33.33	Quinn	92.68	95.00	93.83
Rammus	100.00	100.00	100.00	RekSai	100.00	100.00	100.00	Rell	95.65	92.30	93.94	Renata	90.28	91.39	90.84
Rumble	96.96	96.54	96.75	Ryze	93.10	95.29	94.19	Samira	84.81	83.75	84.28	Sejuncti	95.08	93.61	94.34
Seraphine	92.45	79.67	85.59	Sett	100.00	100.00	100.00	Shen	100.00	100.00	100.00	Shyvana	92.73	94.44	93.58
Sion	93.75	94.49	94.12	Sivir	90.62	87.88	89.23	Skarner	92.50	70.14	79.78	Smolder	95.99	89.33	92.54
Soraka	96.00	100.00	97.96	Swain	81.67	81.67	81.67	Sylas	89.91	88.21	89.05	Syndra	95.50	98.15	96.80
Taliyah	95.61	93.10	94.34	Taric	100.00	100.00	100.00	Thresh	100.00	98.63	99.31	Tristana	96.09	90.36	93.14
Tryndamere	40.00	40.00	40.00	TwistedFate	95.26	83.41	88.94	Twitch	90.36	90.36	90.36	Udyr	98.50	92.25	95.27
Urgot	100.00	100.00	100.00	Vanus	96.00	96.43	96.21	Vayne	97.73	97.73	97.73	Veigar	95.62	95.62	95.62
Viego	93.44	90.16	91.77	Viktor	88.89	87.43	88.15	Vladimir	100.00	100.00	100.00	Voidgrub	0.00	0.00	0.00
Xayah	95.12	94.03	94.57	Xerath	85.29	85.29	85.29	XinZhao	95.28	91.47	93.33	Yasuo	93.88	93.88	93.88
Yuumi	81.17	81.17	81.17	Zac	95.12	97.50	96.30	Zeri	88.79	86.19	87.47	Ziggs	96.39	91.95	94.12
Zoe	87.50	87.50	87.50	Zyra	90.83	76.74	83.19								